Ayush Tiwari

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Summary

Full-Stack Developer with 6 months of production experience building production applications and **AI-powered** applications, real-time industrial dashboards, and scalable data systems. Experienced in delivering end-to-end software solutions across the **SDLC**, plus **DevOps** expertise across CI/CD, containers, and cloud.

Experience

Full stack developer Intern | **THINGSI** | Apr 2026 – Aug 2026

- Optimized **PostgreSQL** databases using materialized views and SQL functions, reducing CPU usage from **98% to 20–25%**.
- Built and deployed optimized **Grafana** dashboards for real-time monitoring, energy reporting, and production tracking across multiple manufacturing plants.
- Developed multiple **Full Stack Projects** and **AI-powered** models for client operations, including OCR-based object detection, **STT** (Speech-to-Text), and **TTS** (Text-to-Speech) voice agents.

JAVA Full Stack Development Intern (Virtual) | **AICTE** | Jul 2024 – Aug 2024

- Built RESTful APIs and backend services using **Java** and **Spring Boot**, with **MySQL** CRUD operations.
- Developed full-stack applications using **React.js**, HTML, and CSS while working in an **Agile** environment with Git/GitHub.
- Implemented **unit testing, debugging**, and **AWS Cloud deployment** for application delivery.

Projects

WebIntel AI (Enterprise Web Intelligence Platform) Frontend | Backend

- Built a fault-tolerant asynchronous web scraping and data extraction engine using **FastAPI**, **Playwright**, and **Crawl4AI**, achieving **96%+** crawl success rate.
- Developed a **Retrieval-Augmented Generation (RAG)** pipeline using **ChromaDB**, vector embeddings, semantic search, and context retrieval to generate relevant, context-aware AI insights from scraped web content.
- Built a real-time **Next.js** and **Recharts** analytics dashboard; optimized **PostgreSQL**, asynchronous **SQLAlchemy**, and data pipelines for crawl monitoring, error diagnostics, and structured AI data processing.

AI-Powered Video Surveillance System Frontend | Backend

- Built a real-time **AI-powered video surveillance platform** using **FastAPI, React.js, WebSockets, OpenCV, YOLO, OCR, and PostgreSQL**.
- Implemented **5 computer-vision capabilities**: vehicle, number plate, fire, crowd, and violence detection with **98%+ detection accuracy**, object tracking, and OCR.
- Engineered **concurrent video processing**, REST APIs, WebSocket streaming, event-driven alerts, and persistent **PostgreSQL/SQLAlchemy** data pipelines.
- Designed a scalable **frontend-backend architecture** for real-time monitoring, automated detection, and surveillance event management.

Paytm Clone (UPI Transaction App) GitHub

- Built a secure UPI-based transaction system with **JWT authentication**, optimized APIs, and scalable database operations.
- **Tech Stack: Next.js**, **TypeScript**, **Node.js**, **Express.js**, **Prisma ORM**, **Tailwind CSS**.

Education & Certifications

B.Tech, Computer Science and Engineering | **Galgotias University** | Nov 2022 – Aug 2026

Certifications: Java Development – Eduskills (IIT Delhi, AICTE) | Software Engineering – NPTEL | **250+ DSA Problems** on **LeetCode** and **Codeforces** (Java)

Technical Skills / Interests

Languages: Python, Java, JavaScript, TypeScript, SQL
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS
Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets , Microservices, Postman
Databases: PostgreSQL (materialized views, query optimization), MongoDB, Prisma ORM, Redis
Data & Visualization: Grafana, ETL pipelines, data systems, Recharts
DevOps & Cloud: Docker, Kubernetes, AWS, GCP, CI/CD, Git, Linux, Grafana, Prometheus